Do Consistent Learned Forces Find the Right Molecular Shape?

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Abstract

A molecular force model can make accurate predictions on stored examples yet lead a molecule to the wrong shape when its forces are followed. We test whether training directly predicted forces to agree with a learned energy improves correct-shape recovery at comparable force error. On aspirin from Landscape17, we implemented energy-derived, direct and consistency-trained E3x models and an independently scored MACE reference. Two-seed validation force errors are comparable for direct prediction and consistency strength 0.1. However, all five checkpoints evaluated with our fixed relaxation procedure recover zero of ten branch references. Follow-up diagnostics expose both reference-force mismatch and extreme accepted coordinate steps. The hypothesis is unsupported under this procedure; its broader validity remains unresolved.

1 Motivation and objective

Quantum-chemistry calculations provide reference energies and forces, but repeating them during simulation is expensive. Learned interatomic potentials offer a cheaper approximation. Their usefulness depends on the motion they produce, not only on error against stored examples.

Energy-derived models obtain forces from one learned scalar energy. Direct models predict vectors, which may be cheaper but need not follow an energy surface. We ask whether encouraging their agreement improves reference-conformation recovery from withheld pathways at comparable force error and cost. A force-converged direct endpoint alone is not an energy minimum; our recovery criterion requires both force convergence and structural agreement.

2 Related work

Landscape17 provides stationary points and descent pathways for evaluating learned potential-energy landscapes [1]. Its results motivate measuring structural recovery alongside predictive accuracy; our withheld-connection study is not a reproduction of its complete kinetic-network benchmark. Bigi et al. identify difficulties with nonconservative forces in geometry optimization and dynamics [2]. This motivates separating force convergence from structural success.

We use E3x to implement a common equivariant backbone [3], and MACE as an established external predictive reference [4]. Physics-informed losses have already been studied for interatomic potentials, including conservative-force constraints [5]. Our contribution is the controlled error–cost–recovery comparison and its failure analysis, rather than a claim that energy–force consistency training is new.

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3 Data, models and evaluation

3.1 Connection-level preparation

We use aspirin from Landscape17, with 21 atoms per structure and hybrid-DFT energy/force labels [1]. Both branches of each transition-state connection stay together. The fixed split contains 26 training, five validation and six test connections, with 4,891, 974 and 1,127 eligible records respectively. Normalization uses training data only. Duplicate/shared structures and reference minima were audited; connections can share minima, so frame count overstates independent support.

Branch destinations come from empirical endpoint classifications, not the ordering of filename pairs. That distinction corrected 26 of 74 branch targets that would otherwise be assigned incorrectly. Structural matching minimizes all-atom RMSD over permitted symmetry mappings and alignment. No trained model has been scored on the test partition. Earlier model-free software fixtures did parse test coordinates and labels for structural checks; they did not select models or solver settings. This distinction is retained in the evidence record.

3.2 Shared architecture and training objective

The E3x models take atomic numbers and coordinates and produce a molecular scalar energy and atomic force vectors. The common backbone has 64 features, two interactions, maximum degree one, eight radial functions and a 5 Å cutoff. Energy-derived forces are $\widehat{\mathbf{F}}^E = -\nabla_{\mathbf{R}}\widehat{E}$. The direct and consistency arms share an auxiliary supervised energy head and a vector force head; ordinary direct prediction is not a force-only fit. The energy model has 43,521 parameters; the dual-head models have 43,649.

With training-derived scales s_E, s_F, we minimize

$$\mathcal{L} = w_E \ \mathrm{mean}\left[\left(\frac{\widehat{E} - E}{s_E}\right)^2\right] + w_F \ \mathrm{mean}\left[\left(\frac{\widehat{\mathbf{F}} - \mathbf{F}}{s_F}\right)^2\right] + \lambda \ \mathrm{mean}\left[\left(\frac{\widehat{\mathbf{F}}^D + \nabla\widehat{E}}{s_F}\right)^2\right]. \quad (1)$$

The final term is present only for consistency training; direct prediction uses $\lambda = 0$. The selected recipe uses $w_E = 0.014584365670390719$ and $w_F = 1.9854156343296092$. All final-setting E3x fits use plain Adam, learning rate 0.003, float64, batch four and 200,000 updates on CPU. Sampling is with replacement: 800,000 sampled exposures correspond to about 163.57 dataset passes. Validation runs every 500 updates, including update zero. BEST is the first minimum validation-force checkpoint, not the final update or a recovery-selected checkpoint.

3.3 Predictive and recovery metrics

The primary force error averages component MAE equally across the five validation connections. For connection g with n_g records and $N = 21$ atoms,

$$\mathrm{MAE}_F = \frac{1}{5}\sum_{g=1}^{5}\frac{1}{3Nn_g}\sum_{r\in g}\sum_{i=1}^{N}\sum_{\alpha=1}^{3}|\widehat{F}_{ri\alpha} - F_{ri\alpha}|. \quad (2)$$

Pooled MAE weights connections by record count and is reported separately. Energy error is molecular residual-energy MAE. Recovery requires maximum absolute Cartesian force component at most 0.01 $\mathrm{eV/Å}$ and symmetry-aware RMSD at most 0.05 Å to the branch's own reference. Converged wrong geometries and nonconvergence remain separate outcomes; the unit of comparison is the whole connection.

4 Predictive results and experimental iterations

Our early 1,000-update energy/direct engineering fits both selected update zero as BEST: force error worsened despite improved energy error. These unsuccessful runs motivated a larger, predeclared budget and a three-configuration baseline comparison. A used width 32 and learning rate 0.001; B used width 64 and 0.003, both with equal normalized loss weights. C retained B's architecture/rate and changed only the supervised weights to the values above. All three completed 200,000 updates.

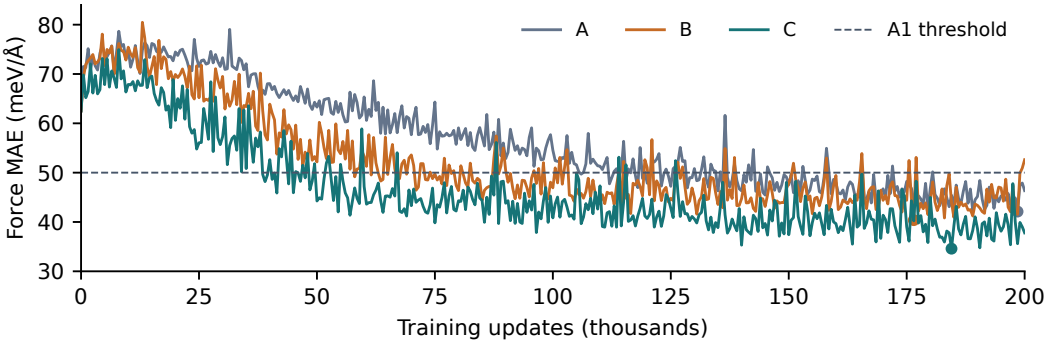


Figure 1: Original energy-baseline validation trajectories, seed 0. Dots mark force-selected BEST checkpoints; the dashed line is the predeclared 50 meV/Å adequacy threshold A1. Each point uses all 974 validation records and equal connection weighting.

C achieved 34.587 meV/Å at update 184,500, compared with B's 40.370 and A's 42.108. It passed A1 and reduced the label-only zero-force reference of 62.504 meV/Å by 44.664%. Its force improvement over B comes with higher energy MAE, 126.990 versus 28.178 meV/molecule, at their separately force-selected checkpoints. A versus B changes both width and learning rate. These comparisons do not isolate width, fixed-update effects or cross-seed repeatability; selection and A1 use the same validation connections.

We then fixed C's recipe for nine subsequent fits: energy seed 1, two direct seeds and two seeds for each $\lambda \in \{0.1, 1, 10\}$. Reusing C seed 0 gives ten final-setting fits in total. Table 1 reports both seed variation and the two-seed means.

Formulation	Force MAE, mean meV/Å	Seed min–max	Energy MAE, mean meV/molecule
Energy-derived	34.893	34.587–35.198	128.725
Direct	38.186	37.291–39.082	76.684
Consistency $\lambda = 0.1$	38.024	36.773–39.275	87.006
Consistency $\lambda = 1$	34.683	33.723–35.643	112.341
Consistency $\lambda = 10$	39.725	39.060–40.390	161.662

Table 1: Validation scores at force-selected BEST. The seed range is descriptive, not a confidence interval. Energy errors use the same saved checkpoints.

Only $\lambda = 0.1$ satisfies the original matching rule: the gap between two-seed mean force errors must be within 10% of the smaller mean and within each arm's seed range. This matches aggregate means, not each seed individually. MACE Run A, trained for 2,048 epochs, independently reproduces 5.349 meV/Å equal-connection MAE (5.260 pooled; RMSE 26.410). Different architecture, training exposure and hardware make it an external reference, not a controlled efficiency comparison.

5 Recovery failure and diagnosis

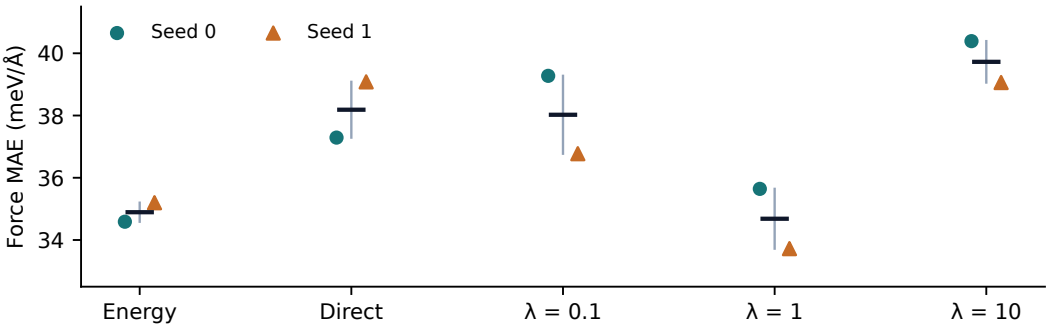


Figure 2: Validation force errors for two seeds per formulation. Ticks mark means; vertical lines join seed extrema, not confidence bounds. Two seeds do not establish superiority.

Consistency strength 0.1 sharply lowers finite-record force/auxiliary-gradient disagreement. However, its mean energy error is higher than direct prediction, and its observed fit-plus-score elapsed time is 1.985 times the direct arm's. This excludes separate diagnostics, earlier failures and unmeasured coordination overhead; it is not total project cost or inference speed. Better agreement on finite validation records does not establish global conservativity.

We evaluated five specific checkpoints: energy C seed 0, direct seeds 0/1 and matched-consistency seeds 0/1. Each starts from the same ten first stored branch frames across five validation connections. No-relaxation own-reference matching is 0/10; every model also recovers 0/10. The recovery pilot gate A2 fails. Across the 50 model/branch outcomes, 25 force-converged endpoints have wrong geometry and the other 25 do not converge. Reusing the same five connections across models does not create 50 independent observations.

The fixed procedure proposes $\mathbf{R}_{\text{trial}} = \mathbf{R} + \eta\mathbf{F}$, initially $\eta = 0.2$ Å^2/eV. Accepted-step growth doubles η up to 1.0; backtracking halves it, with a 10^{-8} floor and at most 40 trials. Each branch has 1,500 force attempts. Trial convergence is checked first; otherwise force agreement permits acceptance when $\mathbf{F}_{\text{trial}} \cdot \mathbf{F}_{\text{known}} \geq -10^{-12}$. There is no energy-decrease test. This caps scale, not displacement, despite inaccurate historical docstring wording. We retained the source and original results.

Checkpoint	Original branch recovery	Reference-start control
Energy C seed 0	0/10	4/6
Direct seed 0	0/10	1/6
Direct seed 1	0/10	0/6
Consistency 0.1 seed 0	0/10	0/6
Consistency 0.1 seed 1	0/10	0/6

Table 2: Post-hoc controls start at six shared reference classes seen in training. They do not change the original branch outcomes or A2.

A 155-evaluation diagnostic reproduced all 50 stopping flags. Every model exceeded the force criterion at all 11 exact DFT minima, whose label forces remained below 2.745 meV/Å. This does not rule out nearby points satisfying our tolerances. All 23 converged dual-head endpoints had disconnected 5 Å neighbor graphs; seven had all atoms isolated and exactly zero direct force. Both converged energy endpoints remained connected, but a separate nonconverged energy endpoint also separated.

Four selected traces reproduced saved endpoints exactly, with largest accepted atom jumps of 436.721 Å (direct) and 268.047 Å (energy). They expose a procedure limitation, without proving sole causality, chemical dissociation or population rates.

6 Discussion and remaining work

The main comparative hypothesis is unsupported under the current procedure: matching validation force errors and improving finite-record energy–force agreement did not produce recovery on these starts. The broader question remains unresolved because all arms, including the energy-derived baseline, failed the original recovery pilot. The diagnostics identify coupled model/procedure limitations rather than a penalty-specific cause. In the strict all-isolated limit of these E3x models, the consistency residual is uninformative; auxiliary gradients at the seven isolated endpoints were not measured.

The 30 reference-start controls above are operational successes/failures, not evidence of exact stationary points, stable minima or absence of nearby solutions.

Our evidence is limited to aspirin, five validation connections and two seeds per final formulation. Validation also selects hyperparameters and BEST checkpoints, so it cannot establish test generalization. Shared minima further limit independence. A stopped direct-force field need not be an energy minimum, and no stability, reaction-rate or barrier claim follows. MACE's predictive quality is verified, while its recovery remains unmeasured.

Changes from the proposal. The submitted proposal planned three seeds and a separate paracetamol replication. We have completed two seeds for the selected E3x settings and deferred paracetamol. A revised adequacy protocol was adopted before the controlled comparison after unsuccessful small-subset checks; the original failures remain documented. Closed-loop work, matched wall-time comparisons and complete inference-cost measurements are still pending. We report these departures explicitly rather than treat planned experiments as completed evidence.

Three next actions. First, finish the MACE checkpoint loader and verify predictions against the saved validation outputs, then run the unchanged ten-start/six-reference procedure control. Second, use that control to decide whether a separately declared solver revision is justified; apply any revision to the same checkpoints and starts and label reused-validation analyses as post-hoc. Third, finish physical/cost diagnostics and freeze the final analysis before any one-time test evaluation. None of these steps requires retraining the current checkpoints.

Team roles and repository. Shedrack completed data preparation, model/training integration, the first MACE reference and current experiments. Olufunke authored the shared recovery evaluator. Emmanuel owns the remaining physical/cost analysis, and Idris owns MACE loading/admission and independent reproduction; these assignments are not completed-result claims. Members will confirm their writing and contribution records before submission. Code and evidence: `https://github.com/Shedrackeze002/idl-molecular-force-consistency`.

References

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